Essays

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Describe how DNA directs the synthesis of salivary amylase. (12 marks)

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Transcription (Max. 4)

- An RNA polymerase molecule binds to the DNA at the beginning of the gene that controls the production of salivary amylase. (1)
- Hydrogen bonds between the bases of the two DNA strands break and the DNA double helix unwinds. (1)
- As the RNA polymerase molecule moves along the DNA template strand, free RNA nucleotides are attached to the template strand through complementary base pairing. (1)
- RNA polymerase catalyses the joining of adjacent RNA nucleotides to form an mRNA. (1)
- Once the whole gene has been transcribed, the mRNA separates from the DNA template strand. (1)
- The mRNA leaves the nucleus through the nuclear pores and enters the cytoplasm. (1)

Translation (Max. 5)

- The mRNA binds to a ribosome in the cytoplasm. (1)
- Each tRNA molecule with a particular anticodon only carries a specific amino acid. (1)
- tRNA with anticodons complementary to the codons of the mRNA will attach to the ribosome. (1)
- Adjacent amino acids are joined together by peptide bonds. (1)
- The ribosome moves one codon at a time down the mRNA, adding one amino acid at a time to the growing polypeptide chain. (1)
- When the ribosome reaches a stop codon, the translation process is complete and the polypeptide formed is released from the ribosome. (1)
- The polypeptide coils and folds to form the protein salivary amylase. (1)

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Applying concepts	9
Communication	3

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